

```
package unit2.mypackage1;
public class Addition {
    public int add(int a,int b) {
        return a+b;
    }
}
```

```
package unit2;
import java.util.*;
public class task01 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size: ");
        int n = sc.nextInt();
        int[] a = new int[n];
        int sum = 0;
        for (int i = 0; i < n; i++) {
            a[i] = sc.nextInt();
            sum += a[i];
        }
        System.out.println("Sum = " + sum);
    }
}
```

```
package unit2;
import java.util.*;
public class task02 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Rows and columns: ");
        int r = sc.nextInt(), c = sc.nextInt();
        int[][] a = new int[r][c], b = new int[r][c];
        for (int i=0;i<r;i++)
            for (int j=0;j<c;j++) a[i][j]=sc.nextInt();
        for (int i=0;i<r;i++)
            for (int j=0;j<c;j++) b[i][j]=sc.nextInt();
        System.out.println("Result:");
        for (int i=0;i<r;i++) {
            for (int j=0;j<c;j++)
                System.out.print((a[i][j]+b[i][j])+" ");
            System.out.println();
        }
    }
}
```

```
package unit2;
import java.util.*;
public class task03 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n=sc.nextInt(), max=Integer.MIN_VALUE;
        int[] a=new int[n];
        for(int i=0;i<n;i++) a[i]=sc.nextInt();
        for(int x:a) if(x>max) max=x;
        System.out.println("Largest = "+max);
    }
}
```

```
package unit2;
import java.util.*;
public class task04 {
    static void findSum(int[] arr) {
        int sum=0;
        for(int x:arr) sum+=x;
        System.out.println("Sum = "+sum);
    }
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] a=new int[n];
        for(int i=0;i<n;i++) a[i]=sc.nextInt();
        findSum(a);
    }
}
```

```
package unit2;
import java.util.*;
public class task05 {
    static void countEvenOdd(int[] arr) {
        int even=0,odd=0;
        for(int x:arr) {
            if(x%2==0) even++;
            else odd++;
        }
        System.out.println("Even = "+even);
        System.out.println("Odd = "+odd);
    }
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] a=new int[n];
        for(int i=0;i<n;i++) a[i]=sc.nextInt();
        countEvenOdd(a);
    }
}
```

```
package unit2;
import java.util.*;
public class task06 {
    static int[] reverseArray(int[] arr) {
        int[] r=new int[arr.length];
        for(int i=0;i<arr.length;i++) r[i]=arr[arr.length-1-i];
        return r;
    }
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] a=new int[n];
        for(int i=0;i<n;i++) a[i]=sc.nextInt();
        System.out.println(Arrays.toString(reverseArray(a)));
    }
}
```

```
package unit2;
import java.util.*;
public class task07 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int r=sc.nextInt(),c=sc.nextInt();
        int[][] a=new int[r][c],b=new int[r][c];
        for(int i=0;i<r;i++) for(int j=0;j<c;j++) a[i][j]=sc.nextInt();
        for(int i=0;i<r;i++) for(int j=0;j<c;j++) b[i][j]=sc.nextInt();
        for(int i=0;i<r;i++) {
            for(int j=0;j<c;j++) System.out.print((a[i][j]+b[i][j])+" ");
            System.out.println();
        }
    }
}
```

```
package unit2;
import java.util.*;
public class task08 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int r1=sc.nextInt(),c1=sc.nextInt();
        int[][] a=new int[r1][c1];
        for(int i=0;i<r1;i++) for(int j=0;j<c1;j++) a[i][j]=sc.nextInt();

        int r2=sc.nextInt(),c2=sc.nextInt();
        int[][] b=new int[r2][c2];
        for(int i=0;i<r2;i++) for(int j=0;j<c2;j++) b[i][j]=sc.nextInt();

        if(c1!=r2) { System.out.println("Multiplication not possible"); return; }

        int[][] p=new int[r1][c2];
        for(int i=0;i<r1;i++)
            for(int j=0;j<c2;j++)
                for(int k=0;k<c1;k++)
                    p[i][j]+=a[i][k]*b[k][j];

        for(int[] row:p) {
            for(int x:row) System.out.print(x+" ");
            System.out.println();
        }
    }
}
```

```
package unit2;
import java.util.*;
public class task09 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int r=sc.nextInt(),c=sc.nextInt();
        int[][] a=new int[r][c];
        for(int i=0;i<r;i++) for(int j=0;j<c;j++) a[i][j]=sc.nextInt();
        for(int j=0;j<c;j++) {
            for(int i=0;i<r;i++) System.out.print(a[i][j]+" ");
            System.out.println();
        }
    }
}
```

```
package unit2;
import java.util.*;
public class task10 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt(),m=sc.nextInt();
        int[][] marks=new int[n][m];
        int[] total=new int[n];

        for(int i=0;i<n;i++)
            for(int j=0;j<m;j++) {
                marks[i][j]=sc.nextInt();
                total[i]+=marks[i][j];
            }

        System.out.println("Marks:");
        for(int[] row:marks) {
            for(int x:row) System.out.print(x+" ");
            System.out.println();
        }

        System.out.println("Totals:");
        for(int x:total) System.out.println(x);

        int highest=0;
        for(int x:total) if(x>highest) highest=x;
        System.out.println("Highest = "+highest);
    }
}
```

```
package unit2;
import java.util.*;
public class task11 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        String s=sc.nextLine();
        String clean=s.toLowerCase().replaceAll("\\s+", "");
        String rev=new StringBuilder(clean).reverse().toString();
        System.out.println(clean.equals(rev) ? "Palindrome" : "Not Palindrome");
    }
}
```

```
package unit2;
import unit2.mypackage1.Addition;
public class task12 {
    public static void main(String[] args) {
        Addition a=new Addition();
        System.out.println("Sum = "+a.add(10,20));
    }
}
```

```
package unit2;
/**
 * Demonstrates Javadoc comments.
 * @author Tharun
 * @version 1.0
 */
public class task13 {
    /**
     * Adds two integers.
     * @param a first number
     * @param b second number
     * @return sum
     */
    public int add(int a, int b) {
        return a+b;
    }

    public static void main(String[] args) {
        task13 t=new task13();
        System.out.println("Sum = "+t.add(10,20));
    }
}
```

```
package unit2;
class Vehicle {
    void start() { System.out.println("Vehicle started"); }
}
class Car extends Vehicle {
    void drive() { System.out.println("Car is driving"); }
}
class SportsCar extends Car {
    void turbo() { System.out.println("Turbo mode activated"); }
}
public class task14 {
    public static void main(String[] args) {
        SportsCar s=new SportsCar();
        s.start();
        s.drive();
        s.turbo();
    }
}
```

```
package unit2;
class Shape {
    void showShape() { System.out.println("Shape"); }
}
class Polygon extends Shape {
    void showPolygon() { System.out.println("Polygon"); }
}
class Triangle extends Polygon {
    void showTriangle() { System.out.println("Triangle"); }
}
public class task15 {
    public static void main(String[] args) {
        Triangle t=new Triangle();
        t.showShape();
        t.showPolygon();
        t.showTriangle();
    }
}
```

```
package unit2;
class Parent {
    void display() { System.out.println("Parent"); }
}
class ChildA extends Parent {
    @Override void display() { System.out.println("Child A"); }
}
class ChildB extends Parent {
    @Override void display() { System.out.println("Child B"); }
}
public class task16 {
    public static void main(String[] args) {
        Parent p;
        p=new ChildA();
        p.display();
        p=new ChildB();
        p.display();
    }
}
```

```
package unit2;
class Constants {
    static final double PI=3.14159;
    final void display() {
        System.out.println("PI = "+PI);
    }
}

public class task17 {
    public static void main(String[] args) {
        Constants c=new Constants();
        c.display();
        // c.PI=3.14; // ERROR: final variable cannot be reassigned
        // class Test extends Constants {
        //     void display() {} // ERROR: final method cannot be overridden
        // }
    }
}
```

```
package unit2;
abstract class Employee {
    String name;
    Employee(String name) { this.name=name; }
    abstract double calculateSalary();
}
class Manager extends Employee {
    Manager(String name) { super(name); }
    double calculateSalary() { return 75000; }
}
class Developer extends Employee {
    Developer(String name) { super(name); }
    double calculateSalary() { return 60000; }
}
public class task18 {
    public static void main(String[] args) {
        Employee e1=new Manager("Anita");
        Employee e2=new Developer("Karthik");
        System.out.println(e1.name+" Salary: "+e1.calculateSalary());
        System.out.println(e2.name+" Salary: "+e2.calculateSalary());
    }
}
```

```
package unit2;
import java.util.*;
public class task19 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt(),b=sc.nextInt();
        try {
            System.out.println("Result = "+(a/b));
        } catch (ArithmeticException e) {
            System.out.println("ArithmeticException: Cannot divide by zero");
        }
    }
}
```

```
package unit2;
class InvalidAgeException extends Exception {
    InvalidAgeException(String msg) { super(msg); }
}
class NegativeSalaryException extends RuntimeException {
    NegativeSalaryException(String msg) { super(msg); }
}
public class task20 {
    static void checkAge(int age) throws InvalidAgeException {
        if(age<18) throw new InvalidAgeException("Age must be 18 or above");
    }
    static void checkSalary(double salary) {
        if(salary<0) throw new NegativeSalaryException("Salary cannot be negative");
    }
    public static void main(String[] args) {
        try {
            checkAge(15);
        } catch(InvalidAgeException e) {
            System.out.println(e.getMessage());
        }
        try {
            checkSalary(-5000);
        } catch(NegativeSalaryException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

```
package unit2;
import java.util.*;
class ZeroDivisorException extends Exception {
    ZeroDivisorException(String msg) { super(msg); }
}
public class task21 {
    static int divide(int a,int b) throws ZeroDivisorException {
        if(b==0) throw new ZeroDivisorException("Divisor cannot be zero");
        return a/b;
    }
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt(),b=sc.nextInt();
        try {
            System.out.println("Result = "+divide(a,b));
        } catch(ZeroDivisorException e) {
            System.out.println(e.getMessage());
        } finally {
            System.out.println("Operation complete");
        }
    }
}
```

```
package unit2;
import java.util.*;
class Book {
    String isbn,title;
    Book(String isbn,String title) {
        this.isbn=isbn; this.title=title;
    }
    @Override public String toString() {
        return isbn+" - "+title;
    }
    @Override public boolean equals(Object o) {
        if(this==o) return true;
        if(!(o instanceof Book)) return false;
        return isbn.equals(((Book)o).isbn);
    }
    @Override public int hashCode() {
        return isbn.hashCode();
    }
}
public class task22 {
    public static void main(String[] args) {
        HashSet<Book> books=new HashSet<>();
        books.add(new Book("111","Java"));
        books.add(new Book("111","Java Programming"));
        books.add(new Book("222","Python"));
        System.out.println(books);
    }
}
```

```
package unit2;
import java.lang.reflect.*;
class Student {
    private String name="Divya";
    private int age=20;
    private void displayInfo() {
        System.out.println("Name: "+name+", Age: "+age);
    }
}

public class task23 {
    public static void main(String[] args) throws Exception {
        Student s=new Student();
        Class<?> c=s.getClass();

        System.out.println("Fields:");
        for(Field f:c.getDeclaredFields()) {
            f.setAccessible(true);
            System.out.println(f.getName()+" = "+f.get(s));
        }

        System.out.println("Methods:");
        for(Method m:c.getDeclaredMethods())
            System.out.println(m.getName());

        Method m=c.getDeclaredMethod("displayInfo");
        m.setAccessible(true);
        m.invoke(s);
    }
}
```

```
package unit2;
interface Animal {
    void sound();
}
interface Pet extends Animal {
    void play();
}
class Dog implements Pet {
    public void sound() { System.out.println("Dog barks"); }
    public void play() { System.out.println("Dog is playing"); }
}
public class task24 {
    public static void main(String[] args) {
        Dog d=new Dog();
        d.sound();
        d.play();
    }
}
```

```
package unit2;
interface Vehicle2 {
    void start();
}
interface Car2 extends Vehicle2 {
    void stop();
    void displayDetails();
}
class Honda implements Car2 {
    public void start() { System.out.println("Honda started"); }
    public void stop() { System.out.println("Honda stopped"); }
    public void displayDetails() {
        System.out.println("Brand: Honda");
        System.out.println("Type: Car");
    }
}
public class task25 {
    public static void main(String[] args) {
        Honda h=new Honda();
        h.start();
        h.displayDetails();
        h.stop();
    }
}
```

```
package unit2;
import java.util.*;
class BookClone implements Cloneable {
    String title;
    ArrayList<String> chapters;

    BookClone(String title,ArrayList<String> chapters) {
        this.title=title;
        this.chapters=chapters;
    }

    public BookClone clone() throws CloneNotSupportedException {
        return (BookClone) super.clone();
    }

    public BookClone deepClone() throws CloneNotSupportedException {
        BookClone b=(BookClone) super.clone();
        b.chapters=new ArrayList<>(chapters);
        return b;
    }
}

public class task26 {
    public static void main(String[] args) throws Exception {
        ArrayList<String> list=new ArrayList<>();
        list.add("Introduction");

        BookClone original=new BookClone("Java",list);

        BookClone shallow=original.clone();
        BookClone deep=original.deepClone();

        shallow.chapters.add("OOP");
        System.out.println("Original after shallow clone: "+original.chapters);

        deep.chapters.add("Exceptions");
        System.out.println("Original after deep clone: "+original.chapters);
        System.out.println("Deep clone: "+deep.chapters);
    }
}
```

```
package unit2;
class Account {
    int accountNumber=101;
    String customerName="Ravi";
    double balance=5000;

    class Transaction {
        void display() {
            System.out.println("Account: "+accountNumber);
            System.out.println("Customer: "+customerName);
            System.out.println("Balance: "+balance);
            System.out.println("Transaction: Deposit");
        }
    }
}

public class task27 {
    public static void main(String[] args) {
        Account a=new Account();
        Account.Transaction t=a.new Transaction();
        t.display();
    }
}
```

```
package unit2;
class Bank {
    static String bankName="ABC Bank";

    static class Branch {
        void display() {
            System.out.println("Bank: "+bankName);
            System.out.println("Branch: Chennai");
        }
    }
}

public class task28 {
    public static void main(String[] args) {
        Bank.Branch b=new Bank.Branch();
        b.display();
    }
}
```

```
package unit2;
class Bank2 {
    void openAccount() {
        class Customer {
            void display() {
                System.out.println("Customer account opened successfully");
            }
        }
        Customer c=new Customer();
        c.display();
    }
}
public class task29 {
    public static void main(String[] args) {
        new Bank2().openAccount();
    }
}
```

```
package unit2;
abstract class Transaction {
    abstract void processTransaction();
}
public class task30 {
    public static void main(String[] args) {
        Transaction t=new Transaction() {
            void processTransaction() {
                System.out.println("Bank transaction processed");
            }
        };
        t.processTransaction();
    }
}
```

```
package unit2;
import java.lang.reflect.*;
interface Calculator {
    int add(int a,int b);
}
class CalculatorImpl implements Calculator {
    public int add(int a,int b) {
        return a+b;
    }
}
class LoggingHandler implements InvocationHandler {
    private final Object target;
    LoggingHandler(Object target) {
        this.target=target;
    }
    public Object invoke(Object proxy,Method method,Object[] args) throws Throwable {
        long start=System.nanoTime();
        Object result=method.invoke(target,args);
        long end=System.nanoTime();

        System.out.println("Method: "+method.getName());
        System.out.println("Result: "+result);
        System.out.println("Time: "+(end-start)+" ns");
        return result;
    }
}
public class task31 {
    public static void main(String[] args) {
        Calculator real=new CalculatorImpl();

        Calculator proxy=(Calculator)Proxy.newProxyInstance(
            Calculator.class.getClassLoader(),
            new Class<?>[]{Calculator.class},
            new LoggingHandler(real)
        );

        proxy.add(10,20);
    }
}
```

```
package unit2;
import java.io.*;
public class task32 {
    public static void main(String[] args) {
        String source="source.txt";
        String destination="copy.txt";

        try(FileInputStream in=new FileInputStream(source);
            FileOutputStream out=new FileOutputStream(destination)) {

            int b;
            while((b=in.read())!=-1)
                out.write(b);

            System.out.println("File copied successfully");

        } catch(IOException e) {
            System.out.println("Error: "+e.getMessage());
        }
    }
}
```

```
package unit2;
import java.io.*;
public class task33 {
    public static void main(String[] args) {
        String file="data.txt";
        int lines=0,words=0,chars=0;

        try(BufferedReader br=new BufferedReader(new FileReader(file))) {
            String line;
            while((line=br.readLine())!=null) {
                lines++;
                chars+=line.length();
                if(!line.trim().isEmpty())
                    words+=line.trim().split("\\s+").length;
            }

            System.out.println("Lines = "+lines);
            System.out.println("Words = "+words);
            System.out.println("Characters = "+chars);
        } catch(IOException e) {
            System.out.println("Error: "+e.getMessage());
        }
    }
}
```

```
package unit2;
import java.util.*;
public class task34 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Number of subjects: ");
        int n=sc.nextInt();

        int total=0;
        for(int i=0;i<n;i++)
            total+=sc.nextInt();

        double average=(double)total/n;

        System.out.println("Total = "+total);
        System.out.println("Average = "+average);
    }
}
```

```
package unit2;
import java.io.*;
public class task35 {
    public static void main(String[] args) {
        File file=new File("sample.txt");

        try {
            if(file.createNewFile())
                System.out.println("File created");
            else
                System.out.println("File already exists");

            System.out.println("Name: "+file.getName());
            System.out.println("Path: "+file.getAbsolutePath());
            System.out.println("Size: "+file.length()+" bytes");

            if(file.delete())
                System.out.println("File deleted");
        } catch (IOException e) {
            System.out.println("Error: "+e.getMessage());
        }
    }
}
```

```
package unit2;
import java.nio.file.*;
import java.io.*;
public class task36 {
    public static void main(String[] args) {
        Path source=Paths.get("source.txt");
        Path destination=Paths.get("nio_copy.txt");

        try {
            Files.copy(source,destination,StandardCopyOption.REPLACE_EXISTING);
            System.out.println("File copied using NIO");
        } catch (IOException e) {
            System.out.println("Error: "+e.getMessage());
        }
    }
}
```

```
package unit2;
import java.io.*;

class StudentData implements Serializable {
    String name;
    int rollNo;
    transient String password;

    StudentData(String name,int rollNo,String password) {
        this.name=name;
        this.rollNo=rollNo;
        this.password=password;
    }
}

public class task37 {
    public static void main(String[] args) {
        StudentData s1=new StudentData("Kavya",101,"secret");

        try {
            ObjectOutputStream out=
                new ObjectOutputStream(new FileOutputStream("student.ser"));
            out.writeObject(s1);
            out.close();

            ObjectInputStream in=
                new ObjectInputStream(new FileInputStream("student.ser"));
            StudentData s2=(StudentData)in.readObject();
            in.close();

            System.out.println("Name: "+s2.name);
            System.out.println("Roll No: "+s2.rollNo);
            System.out.println("Password: "+s2.password);

        } catch (IOException | ClassNotFoundException e) {
            System.out.println("Error: "+e.getMessage());
        }
    }
}
```